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CHAPTER

8 Spatial Attention

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Abstract

Location plays a fundamental role in visual attention. Experiments using spatial cuing, spatial probes, flanker interference, and other methods show that visual information is often selected by its location. This spatial attention can take the form of a gradient, with facilitation that weakens with distance from the center. Although attention is easily allocated to a cued location, it can also be driven effectively by features such as motion, color, and orientation, and this feature-driven selection is accomplished at least in part by selecting locations with those features. Attention is flexible and often seems to be allocated to objects, as demonstrated in experiments with overlapping objects. Attention to one part of an object can increase attention to other parts of the same object, even though the object organization is irrelevant to the task. More experiments, including neuroscience studies, are necessary to determine the relationships between location-based, feature-based, and object-based attention.

Keywords: [location](#), [selection](#), [cuing](#), [flanker](#), [feature](#), [object](#), [split](#), [gradient](#)

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“Spatial attention” refers to selection that is based on location. In most circumstances, a perceptual system cannot fully process all the incoming information, and therefore some aspect of processing is limited to just information from a specific location or region. One aspect of spatial selection is very obvious in vision: At any given moment, the two eyes are fixated at a particular location in space, allowing the visual information from that location to be processed by the retinal fovea, which provides higher spatial resolution than any other part of the retina. Multiple times each second, the eyes will move to a new location, so that over a relatively brief time the powerful processing resources of the retina will be applied to many different locations in the environment. Thus, eye movements are one example of a limited resource (spatial acuity) and a selection mechanism for choosing how to allocate this resource to different locations. Because eye movements are easily observed, this selection is often described as “overt attention.”

Eye movements, however, are just the beginning of the story of spatial attention in vision. There are other limits on visual processing, and other types of visual selection to compensate for them. Von Helmholtz

(1924) demonstrated “covert” visual attention that selected a specific region of the visual field without moving the eyes (see Van der Heijden, 1992). In a dark room, he fixated his eyes at the center of a large piece of paper on the wall covered with many different letters. He then generated an electric spark that illuminated the room. The flash was too brief for an eye movement, but Helmholtz found that he could attend to and perceive a small set of letters from one area. Before the next flash, he could shift his covert attention to a different location, and once they were illuminated, he could perceive those letters, even though his eyes had not moved.

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Visual System Limitations

Helmholtz’s experiment demonstrates a basic limitation in visual processing, along with a method for coping with the limitation. Helmholtz was unable to identify all the letters in the visual field within the short time interval of the light flash, but he could direct his visual processing resources to a particular location, in order to process the letters there. The limitation in visual capacity is not surprising given the basic structure of the visual system. The initial neural processing stages, from the retina to the lateral geniculate nucleus of the thalamus and on to the occipital regions such as V1 and V2, are organized as large spatial maps in which the neurons have relatively small receptive fields. In other words, each neuron represents information from a small region of space, and the huge collection of neurons working simultaneously together represent the entire visual field. These structures feed into the “What” pathway that identifies visual objects, and in the upper levels of this pathway, the receptive fields become larger and larger. At these levels, each neuron can represent complex shape properties, but because of its large receptive field, it can be driven by multiple objects at different locations. Because information from different objects converges together in the What pathway, there can be much confusion about which properties belong to which object. This confusion can lead to “binding errors” in which features from different objects are incorrectly combined together to form an illusory conjunction. Binding errors can be prevented by a selection mechanism that only allows information from a single location or object to activate these neurons at any given time. Because of the potential for binding errors, the recognition system is effectively a limited-capacity processing resource, and the selection mechanism decides how to allocate this resource.

The early stages of visual processing appear to involve relatively simple neural operations that are performed in parallel across the visual field, while the later stages require more complex operations that can only be applied to a limited part of the input. Because the retina and other early structures are organized as spatial maps, it is perhaps not surprising that location seems to serve as a basis for selection. A huge number of experiments conducted over many years support the idea that visual attention works spatially, by selecting the information from a specific location in the visual field.

Spatial Cuing Experiments

One key piece of evidence that visual attention selects from a spatially organized representation comes from spatial cuing experiments, which have demonstrated that a response to a visual stimulus is faster and/or more accurate when its location is known in advance. In these experiments, the test or probe stimulus to be reported is often preceded by a cue stimulus that indicates a particular location where the probe is likely to appear (Eriksen & Hoffman, 1974; Posner, Nissen, & Ogden, 1978; Posner, Snyder, & Davidson, 1980). After the cue, the eyes remain fixed in these experiments, which is confirmed in some cases either by tracking eye position or by limiting the display time between cue and probe stimulus. Thus, the facilitation at the cued location is due to covert attention and not to the higher spatial resolution at the fovea.

The cuing can be done in a couple of different ways, and the different results suggest two different attentional control mechanisms. The more straightforward approach is to position the cue at or very near the location where the upcoming probe can be expected. Attention is simply allocated to the location with the cue and is there waiting when the probe stimulus appears. This method is called exogenous or peripheral cuing, because the cue is usually located somewhere away from the center of gaze. The alternative is for the cue to be a symbol that appears at the fixated location. This method is known as endogenous or central cuing. Attention is generally allocated more slowly under endogenous cuing than under exogenous cuing, because of the extra step of interpreting the cue in order to determine the location to be attended. Exogenous cues also trigger an attention shift more automatically and are more difficult to ignore than endogenous cues. When a salient cue stimulus appears at a location, it is difficult not to attend to that location, even when it is known that the probe is no more likely to appear at the cued location than anywhere else.

A spatial cue can speed responses in a simple detection task in which no discrimination is necessary and in which few distracting stimuli are present. These strong attention effects from these simple tasks indicate that spatial attention may be playing some role other than the filtering out of interfering distractors, as suggested earlier. One alternative explanation is that spatial cues do not actually influence the effectiveness of visual processing, but only lower the threshold for responding, allowing faster responses but also more errors. However, Downing (1988) measured sensitivity using d' (Macmillan & Creelman, 2005) in a collection of detection and discrimination tasks to show that foreknowledge of the stimulus location actually improved perceptual sensitivity, rather than simply lowering a response threshold.

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Spatial Probe Experiments

In the spatial cuing paradigm, a simple cue causes attention to be allocated to a particular location, and the effects of attention are measured on the perception of a probe stimulus. Probe stimuli can also be used to measure how spatial attention is allocated in the course of other visual tasks. Hoffman and Nelson (1981) combined two different tasks: searching for target letters in an array of rapidly changing letters, and monitoring for a small U-shaped stimulus (the probe) to appear. When subjects were correct in their response for the letter task, they were more likely to also correctly report the orientation of the U-shaped stimulus if it was near the target letter. On the other hand, when their letter response was incorrect, their report of the U-shaped orientation was more likely to be correct if it was not near the target letter. Attention allocated for the letter task also benefited the processing of the U-shaped stimulus when it was at a nearby location. Hoffman, Nelson, and Houck (1983) demonstrated that attention allocated in the detection of a flickering dot could enhance the identification of a nearby probe letter.

These two studies demonstrate the generality of spatial attention: When it is allocated for one task, it can benefit very different stimuli used in a very different task. They also address concerns that location only plays an important role in cuing tasks because the instructions emphasize location. In Hoffman et al.'s experiments, location was not part of the instructions, and yet the attention to the probe stimulus was still determined by its location relative to the stimulus from the other task. Finally, these experiments demonstrate how spatial probes can be used to measure attentional allocation during other visual tasks.

Kim and Cave (1995) combined a visual search task with two different types of spatial probes. In both sets of experiments, a small array of colored shapes was presented so briefly that there was not enough time to execute an eye movement before they disappeared. The search task was to find a red square among a set of red circles, green squares, and green circles. In one set of experiments, the search array was followed on some trials by a small dot at a location formerly occupied by one of the colored shapes. The response time to the dot indicated the degree to which that location had been attended. In the other set of experiments, the

search array was followed by an array of small letters, with one letter at each of the locations formerly occupied by a shape. Subjects identified as many letters as possible, and the letters they reported indicated which locations had been attended. Thus, both methods served as a measure of spatial attention, because the probes differed from the search stimuli in color, shape, and most other properties, and shared only location. The fact that probe responses were affected by attention indicated that attention selected the location occupied by one of the search items, and ended up also selecting the probe that appeared at that location. A similar pattern emerged from both types of probe: Attention was strongest at the location of the red square target, and it was somewhat weaker at locations that had the target color (red) or the target shape (square) but not both. Attention was weakest at the locations of the green circles. Attention was apparently allocated to each location according to the target features present there.

Probe experiments like these are complicated by the fact that subjects must perform the primary and probe tasks together. The goal is to measure how spatial attention is allocated in the performance of a visual search or some other primary task. The measure of attention comes from the probe task. The executive control necessary for the probe task may interfere with the primary task, and the need to report the probe may alter the allocation of attention from what it would normally be for the primary task. Event-related potential (ERP) measures allow for the measurement of spatial attention without a probe response, allowing subjects to focus on performing just the primary task.

Event-Related Potential Experiments

Many ERP attention experiments have used a procedure similar to the dot-probe experiments described earlier. While the subject is engaged in the primary task, a dot probe appears at some location in the display. The probe generates a wave of neural activity that can be measured with electroencephalographic (EEG) electrodes on the scalp. The changes in electrical potential over time can be plotted in a graph. When the changes are plotted relative to the time of the probe appearance, the resulting waveform is an ERP. The size of the changes in electrical potential over time is shaped by attention. When a probe appears at an attended location, the potential from electrodes at certain scalp locations will be more positive in some time windows and more negative in other time windows than they would be without attention to the probed location (Luck, Woodman, & Vogel, 2000; Mangun & Hillyard, 1995). ERP thus provides another method for measuring how attention augments the signals from stimuli at selected locations in the visual field, without requiring that a probe task be combined with the primary task.

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Spotlight Metaphor

The results from these different experimental methods converge on the idea that visual information is selected according to its location. Posner, Snyder, and Davidson (1980) illustrated this idea by comparing visual attention to a spotlight. The attentional spotlight is directed at a single location, and anything that falls within its beam is selected, while everything in the dark regions outside is excluded. Thus, when the beam is directed to a specific location in the course of a visual search, a spatial probe that appears at the same location will be selected and facilitated. The spotlight metaphor has had an enormous effect on theorizing about visual attention, and it has suggested to many that because visual attention shares the spatial aspect with a spotlight, it must share other properties as well, such as its shape and manner of motion. Some of those assumptions have been substantiated, while others have not (Cave & Bichot, 1999).

Attentional Gradient

In addition to demonstrating the spatial nature of visual selection, the spatial cuing, spatial probe, and ERP methods all provide ways for the pattern of attentional facilitation to be mapped across space. Some, but not all, attentional tasks produce a gradient pattern, in which attentional facilitation is strongest in the center and declines gradually with distance from the center. In other words, the spotlight of attention can often have a very fuzzy edge.

Some of the earliest evidence for an attentional gradient came from experiments by Eriksen and colleagues. Eriksen and Eriksen's (1974) subjects simply reported a target letter that appeared in the center of the display, accompanied by flanking distractor letters. Each target letter belonged to one of two categories, and subjects responded by pressing one of two buttons. Even though the location of the target was known in advance and it occurred in the fovea, the distracting flankers had a subtle but measurable effect on the responses. When the flankers were associated with the same response as the target, responses were faster, and when the flankers were associated with the other response, they were slower. The results demonstrate a partial failure in spatial attention. The spatial nature of the selection in this task was demonstrated by moving the flankers more than 1 degree of visual angle away from the target, which eliminated the flanker interference. Attention that was allocated to the target location spilled out into the neighboring regions, where it allowed distractors to partially activate responses.

This spilling over of spatial attention was made more evident in cuing studies by Downing (1988; Downing & Pinker, 1985), which were similar to Posner's cuing experiments except that the target could appear at any of 10 or 12 locations, each marked by a box. When one location was cued as being the most likely location of an upcoming target stimulus, the attentional facilitation for that stimulus was strongest at the cued location, of course, but there was weaker but still measurable facilitation at nearby locations, falling off gradually with the distance between cued location and target location. The gradient findings suggest that spatial attention may not be all that precise; although attention is allocated for one stimulus at one location in the visual field, its effects sometimes spill over to neighboring regions.

This distance effect, in which response time rises with distance between cue and probe stimuli, has been interpreted in different ways. Once a location is cued, attention might be allocated in a gradient pattern, with strong facilitation in the center and weaker facilitation at the edges, and this pattern might stay in place as the probe stimulus appears and is processed. This account assumes that attention can speed processing along, but that a stimulus can still be identified and responded to even if it is at a location with little or no attentional facilitation. Under an alternative account, attention might be focused narrowly and precisely at the cued location, and then moved to the probe location once it appears. In this account, no stimulus can trigger a response until it is attended, making it necessary for attention to move to the location of a stimulus before response. This attentional movement would explain the distance effect in response times, but only if it takes more time to move attention longer distances.

However, there have been a number of arguments against the idea that movement time is proportional to distance (see Cave & Bichot, 1999, for a review). On top of that, LaBerge and Brown (1989) tested these two accounts against one another with a complex probe experiment. The primary stimulus was a target letter in the center of the display with 4 distractors on both sides. It was quickly followed by a probe letter that appeared somewhere along the line of locations previously occupied by the target and distractors and was also accompanied by flanking distractors. Response time increased with the distance between the target and the probe, which could be explained with either a stationary attentional gradient allocated around the target location, or a small attentional spotlight sliding across from the target to the probe location. However, the distance effect was stronger when the probe was very similar to its flanking distractors. LaBerge and Brown argued that similarity should not affect the speed of a sliding spotlight, and so they concluded that there

must be a stationary gradient of attention. There is an additional reason to favor the fixed gradient account over the moving spotlight account: Similar gradient patterns are found with ERP attention measurements (Mangun & Hillyard, 1988).

Surprisingly, not all experiments have produced a gradient pattern. Hughes and Zimba (1985, 1987) used a cuing paradigm similar to Downing and Pinker's but with a sparser stimulus array that had no boxes marking the possible stimulus location. They found uniformly fast responses whenever the cue and the target stimulus were both in the same hemifield, and slower responses when they were in different hemifields. The squares apparently have an important effect on attention in this task, because when Zimba and Hughes (1987) added squares to mark their potential target locations, they found a gradient pattern similar to Downing and Pinker's. Attention can thus take the form of a broad gradient in some circumstances, especially when there are items in the display that may produce a low level of interference. In emptier displays, attention seems to spread even further, constrained only by hemifield boundaries.

The cuing experiments by Downing (1988) mentioned earlier provide additional evidence that the size of the gradient can be adjusted to fit the circumstances. Using a signal detection approach (Macmillan & Creelman, 2005), she measured the falloff in sensitivity with distance from the cued location. Sensitivity fell off more quickly and thus produced a tighter, steeper gradient, when the stimulus items were closer together. Importantly, the shape of the gradient changed with the task, with luminance detection and brightness discrimination producing a broader gradient than orientation and form discrimination.

The flexibility in attentional allocation depending on the task suggests that spatial attention can be more narrowly focused when there is more potential for interference between different objects in a stimulus array. In Downing's experiments, this focusing occurred when there were more items present, when the items were closer together, or the discrimination was more demanding. For all these experiments, having a narrower focus of spatial attention facilitated task performance. In Kim and Cave's search task, in which each target was accompanied by multiple distractors sharing some features with it, there seemed to be little spread of attention from one object to the next, suggesting that spatial attention can be reasonably precise when it needs to be. Cave and Zimmerman (1997) found a similar pattern with a letter search task: All of the distractors were inhibited relative to the target location. Interestingly, although spatial attention was focused in a fairly narrow fashion in this task, it exerted a stronger effect when the distractor letters were more confusable with the target letter, demonstrating again that spatial attention varies according to the demands of the task.

Taken together, these results suggest that in tasks with higher distractor interference, the selected region can be smaller and the strength of the facilitation within this region can increase. Spatial attention is apparently serving to limit distractor interference, but it does not inhibit distractor processing any more than is necessary given the current task. This idea is generally consistent with the assumptions underlying perceptual load theory (Lavie, Hirst, de Fockert, & Viding, 2004), which postulates that more perceptually demanding tasks will require attention to be narrowly focused on a target object, while less demanding tasks will allow some attention to spill over onto distractor objects (but see Benoni & Tsal, 2010).

Attentional Zoom

If spatial attention is as flexible as suggested here, then there needs to be an ability not only to direct attention to a specific location but also to adjust it to select a region of a certain size. Controlling the size of the selected region could also be advantageous in selecting stimuli of various sizes, or when the region within which a stimulus is expected to appear can vary in size over time. Eriksen and St. James (1986) compared attention to a zoom lens with the ability to select smaller or larger regions according to the situation.

LaBerge (1983) provided evidence for adjustments in attentional size using a probe paradigm similar to that from LaBerge and Brown described earlier. There were two conditions, which were tested in separate blocks. In the wide attention condition, the primary stimulus was a five-letter word, and in the narrow attention condition, it was a single letter with two distractor characters on either side. On some trials, the primary stimulus was replaced by a probe stimulus consisting of a single character at one of the five locations normally occupied by one of the characters in the primary stimulus. When the probe response times for the five different positions were compared against one another, they showed that attention was sharply focused in the narrow condition, producing fast response times for probes in the center location and slower response times for those at other locations. In the wide condition, however, attention was more evenly spread across the five conditions, producing more similar response times. Subjects were able to adjust their “attentional zoom” setting according to their expectations for the upcoming task.

The size adjustments demonstrated by LaBerge may or may not be related to a different type of size adjustment demonstrated by Larsen and Bundesen (1978) and by Cave and Kosslyn (1989). In these experiments, a single stimulus appeared by itself on each trial. Many of these stimuli were the same size as the preceding stimulus, so that subjects knew what size to expect. Some stimuli, however, appeared at a different size, and on these trials, the response time was proportional to the change in size from the previous stimulus to the current stimulus. Subjects appeared to prepare themselves to process a stimulus of a cued size, just as subjects in the standard spatial cuing experiments set themselves to process a stimulus at a cued location. In both the size and location experiments, the response time was proportional to the difference between the expected and actual stimulus properties. When these subjects prepare for a particular size, they may be selecting a region of that size, as suggested by LaBerge’s (1983) experiment. On the other hand, the adjustment may not be linked to a particular location but may simply cause recognition mechanisms to expect a stimulus at a certain scale.

Split Attention

In many circumstances, there are two or more important visual objects that are visible simultaneously. Given the ability to zoom attention in and out, it ought to be possible to select an area that is large enough to select all of the target objects. If there are distractor objects in between the target objects, however, then they would be selected as well, which would interfere with processing of the targets. A number of studies have tested whether two different locations can be selected simultaneously without selecting the regions in between. This has been a very difficult question to answer, because most of the experimental results that are predicted by split attention could also be produced by a single area of attention that moves quickly from one location to another, or that occupies one target location on some trials and another location on other trials.

There is only an advantage to splitting attention if there is also processing capacity to identify the two selected objects simultaneously. There is not a clear experimental demonstration of multiple objects being recognized simultaneously, but there is evidence for a generally high degree of visual processing capacity. Some experiments have demonstrated that the analysis of complex scenes is very fast, while others have demonstrated the effects of visual context on recognition of a target object, and still others have measured attention in the execution of tasks that require coordinating movements involving multiple objects (see Cave, Bush, & Taylor, 2010, for a review).

Doubts arose early on about the idea of split attention as a result of a cuing experiment by Posner, Snyder, and Davidson (1980). When they cued a secondary location along with a primary location, the secondary cue had no effect unless it was adjacent to the primary cue location. Posner et al. interpreted this pattern as evidence that the secondary location could only be selected by expanding the spotlight of attention, rather than by splitting it. However, a number of studies since then using methods other than simple spatial cuing

have suggested that attention can be split, especially in circumstances in which the two targets will not interfere much with one another and share some feature such as color that sets them apart from distractors. Jans, Peters, and De Weerd (2010) have reviewed these studies in some detail, and they concluded that none of them definitively demonstrate split attention, although some argue that their criteria are overly stringent (Cave, Bush, & Taylor, 2010).

Distractor Inhibition

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Another way to select multiple target locations is to inhibit only the locations containing distractors, leaving the target locations and blank locations uninhibited. Because of the importance attached to the spotlight metaphor, most studies have been interpreted in terms of processing facilitation occurring at target locations, and this sort of attentional $\searrow$ spotlight seems to be the most straightforward explanation for experiments such as Downing's (1988) showing a gradient of attention. However, in some circumstances the results could be explained by inhibition of distractor locations rather than facilitation of target locations, and Cepeda, Cave, Bichot, and Kim (1998) used spatial probes to show that in a search for a red digit among three green distractor digits, inhibition was focused specifically on the green locations. Thus, the attentional system apparently has the option of targeting inhibition specifically on distractor locations when that approach is beneficial.

The inhibition seen in this experiment may be related to negative priming, in which a stimulus that has been a distractor on one trial produces a weaker response on a later trial (Neill, 1977; Reisberg, Baron, & Kemler, 1980; Tipper, 2001). Negative priming can be associated with specific objects or stimuli, but it can also be very general in that it applies to objects independently of their location. Its underlying mechanisms are a topic of debate, making it difficult to know whether those mechanisms are responsible for the spatial inhibition that is driven by color in Cepeda et al.'s study.

Inhibition of distractor locations has been demonstrated a number of times for distractors that are positioned very close to a target object (e.g., Hopf et al., 2006; Mounts, 2000). This pattern is interesting not only because it demonstrates that attentional selection can work partly through inhibition but also because it suggests that distractors near a target interfere more with the target's processing than distractors farther away. This interference probably arises because the nearby distractors will fall within the receptive fields of many of the visual neurons that will be responding to the target. Inhibiting those distractors will therefore allow those neurons to represent the properties of the target more accurately, rather than the properties of the distractors. The pattern of facilitation at the target location surrounded by a ring of inhibition to block nearby distractors is very similar to the center-surround patterns that trigger neural responses most effectively at many levels within the visual system, starting with the retinal ganglion cells.

Multiple Object Tracking

One task that seems to require split attention is the Multiple Object Tracking (MOT) task. At the beginning of a trial, a set of identical shapes are shown at random locations across the display. A subset of these shapes is then temporarily marked to indicate that they are targets, while the remaining shapes are distractors to be ignored. The marks disappear, and all the objects move across the display, each moving independently of the others. The targets must be tracked throughout this motion, and at the end of the trial, a test determines whether subjects can distinguish between targets and distractors. Typically four objects can be tracked with relatively few errors.

Are there four different spotlights of attention, each tracking a different target as it moves around the display? An experiment by Pylyshyn, Haladjian, King, and Reilly (2008) suggests a different explanation. They combined the MOT task with spatial attention probes similar to those used by Cepeda et al. and found evidence for inhibition that was targeted just at the locations with distractors. Interestingly, when the distractors were not moving or were in a different depth plane from the targets, they were not inhibited. Apparently those distractors were easily excluded by other means, and the distractor inhibition was only used for those distractors that were more easily confused with the targets. Pylyshyn et al. thus provide another piece of evidence that simultaneous attention to multiple objects can be accomplished by distractor inhibition.

Visual Search

As is clear from the descriptions earlier, visual attention research benefits from being able to draw on a variety of different types of experiments. One of the most heavily used methods is visual search, in which a target object appears at an unknown location surrounded by many distractor objects. In many experiments, the task is simply to report whether a target is present in the array, although in some experiments, a target defined by one property (e.g., color) will be present on every trial, and once it is found, another of its properties (such as its shape) must be reported.

There are two aspects to attentional control in most visual search tasks. Usually, the eyes are allowed to move from location to location during the search. This overt attention is by definition spatial, since the goal is to choose the best location for the next fixation. Eyetracking measurements can reveal which locations in the search array are chosen, and thus indirectly indicate which visual properties are guiding overt attention. For instance, color is especially effective at guiding eye movements, and properties such as orientation and shape can also be used effectively (Williams & Reingold, 2001). The effectiveness of a particular feature in guiding overt ↵ attention will depend on the discriminability of the target feature relative to the corresponding distractor features. It is much easier to direct the eyes to a red target among green distractors than a red target among orange distractors.

The other aspect of attentional control in search is harder to measure. Given the evidence described earlier for covert attention shifts, it seems likely that covert attention will shift from one item to another during a single fixation. A number of theories, including Feature Integration Theory (Treisman & Gelade, 1980) have assumed that this covert attention is also selecting by location, although the spatial basis of covert attention is harder to demonstrate than it is for overt attention. The data produced by most visual search experiments are response times and error rates for finding the target, and although these data are useful for gauging the attentional limits on visual processing, they do not indicate whether selection is based on location or something else. One relevant piece of evidence for spatial selection in search comes from Kim and Cave's (1995) spatial probe experiments with a search task in which eye movements are prohibited. Even though the targets are defined by a combination of shape and color, the responses to the spatial probes indicate that the covert selection is ultimately based on location.

If the assumption is correct that covert attention in visual search is selecting by location, there are other questions that arise about how much is selected at any given moment. Many theories (e.g., Treisman & Gelade, 1980) suggest that in difficult searches, one item is selected at a time, with attention moving serially from item to item until the target is found. This claim is supported by the fact that response times in many difficult searches increase linearly with the number of items in the display. However, the linear increase in response times is also consistent with a system that selects a small group of items and processes them simultaneously, and then moves serially from one clump to the next (Pashler, 1987). Given the evidence discussed earlier that "attentional zoom" can be adjusted to select either a small or a large region at any one

time, it seems likely that easier searches might be conducted with a rather wide attentional zoom setting, which selects a large clump of items and processes them together, so that a large display can be processed with just a few attentional shifts. More difficult searches might require a more focused attentional zoom setting, so that fewer items are processed in each step, and more attentional shifts are necessary (see Nakayama & Joseph, 1998).

The flexibility in attentional control makes it difficult to know exactly how spatial attention is used in visual search, and the uncertainty gets worse as we consider this question further. Townsend (1990) has suggested that the fact that response times increase with the number of items in a search display does not necessarily indicate that the search is serial, with attention shifting from location to location. Instead, this pattern might reflect a process that is completely parallel but limited in capacity, so that a fixed amount of processing resources is distributed across all the objects in the search array. All of the items are processed simultaneously, but the processing speed for all the items is slower for large displays than for small displays, because a smaller proportion of the resources is applied to each item. Thus, there are a number of theoretical options for explaining visual search and the role of covert spatial selection in it, all making similar predictions. We know that visual attention must be flexible enough to deal with many different processing challenges, and that flexibility makes it very difficult to pin down.

Selecting From Three-Dimensional Surfaces

By definition, spatial attention is selecting locations, but that definition does not specify what type of location-based representation the information is selected from. One straightforward assumption is that selection occurs within a two-dimensional representation early in the visual system, in which basic features such as color and orientation have been identified, but which has not been processed much beyond that. However, some experiments suggest that visual attention can be guided by information about depth. Nakayama and Silverman (1986) showed that search can be limited to one depth plane, ignoring distractors in other depth planes, just as it can be limited to one side of the display, ignoring distractors on the other side. This result suggests that search occurs within a three-dimensional (3D) rather than a two-dimensional representation. He and Nakayama (1992) extended this result. Their subjects searched for a patch of a specific color among other color patches, all organized into three different depth planes. When the three depth planes were all oriented vertically with respect to the viewer, and the color patches were also vertical so that they lay flat within the planes, search was easy. There was only one patch with the target color within the target depth plane, and subjects were able to limit search to that depth plane and easily find the uniquely colored target. However, when the color patches were all tilted so that they “stuck out” from their depth planes, search could no longer be limited to a single plane, and responses were much slower. The surprising result came when the depth planes were all tilted so that they were no longer vertical. With tilted depth planes, search was still efficient as long as the color patches were aligned with their depth planes, even though that meant that the color patches were now tilted with respect to the viewer. If the color patches were vertical and not aligned with the depth planes, search was difficult. Thus, visual search seems to be operating within a representation of a 3D surface.

Additional evidence for the use of depth information in search comes from Enns and Rensink (1990), who constructed a target pattern of three different four-sided polygons, each with a different level of shading. When this pattern was presented among distractors that were a different arrangement of the three component shapes, search was very difficult. However, when the three components of the target and distractor shapes were rearranged so that they looked like shaded 3D cubes, search became very easy. Once again, the results suggest that a 3D representation is constructed before search begins.

Marr (1982) theorized that visual processing could be broken down into three stages, and that the intermediate stage was devoted to reconstructing the third dimension. He described the representation produced by this stage as a “ $2\frac{1}{2}$ -D sketch.” The “sketch” indicated that it was still a spatially oriented depiction of the scene represented from a specific viewpoint. The “ $2\frac{1}{2}$ -D” indicated that it was midway between the 2D representation from early vision and a full 3D model. The experiments in 3D search suggest that visual search may be conducted within a representation like a $2\frac{1}{2}$ -D sketch, which is organized spatially but includes information about the depth at each point.

Attention Based on Locations or on Features?

The Kim and Cave (1995) study mentioned earlier indicates that when searching for a target defined by shape and color, attention is ultimately allocated to the locations containing the target features. There is plenty of additional evidence that when the task is to select a specific feature such as color, the feature selection is accomplished indirectly by selecting a location (Lamy & Tsal, 2000; Shih & Sperling, 1996; Tsal & Lavie, 1993). Harms and Bundesen (1983) found that when attention is focused on a target object, it spreads to a nearby distractor with the same color, providing another example of selection driven by color, but Kim and Cave (2001) used spatial probes to show that it is ultimately location based. All of these studies are consistent with an attentional mechanism that facilitates processing at any location that has a specified target feature. As long as the facilitation applies to all stimuli at the selected location, including items that do not have the target feature, then this would be considered to be indirect feature selection that is mediated by location selection.

Even though location selection contributes to feature selection in many circumstances, there still might be a separate mechanism for pure feature-based selection. In other words, within the visual system there might be a representation organized by color (or some other feature) that is separate from the representations organized by location, and that color-indexed representation might be used to select information directly according to color, without selecting other items at the same locations. Finding experimental evidence for this direct feature selection, however, has been difficult. For instance, Vierck and Miller (2006) tried to make location irrelevant to a search task by presenting a series of letters one after the other in the center of the screen. This rapid serial visual presentation (RSVP) procedure effectively turns a spatial search into a temporal search, requiring subjects to monitor the stimulus stream over time until the target appears. Their letters appeared in different colors, and when subjects knew the target color in advance, their detection performance improved. This may be an example of direct color selection, but if we consider a location selection mechanism like the one suggested earlier in which the amount of facilitation at each location is controlled by the color at that location, then these results could also be explained by location selection. When a nontarget color appears in the target location, it is less facilitated than when the target color appears.

Although there are questions about the behavioral evidence for direct feature selection, neuroscience studies have provided more convincing evidence. Brain imaging has shown that while subjects are attending to a stimulus with a particular target feature (color or direction of motion) at one location, another stimulus at an unattended location can produce an enhanced neural response if it shares the target feature (Saenz, Buracas & Boynton, 2002; Treue & Martinez-Trujillo, 1999), and Zhang and Luck (2009) have used ERPs to demonstrate that under the right conditions, color selection can shape perceptual processing at a very early stage. ⁴ Thus, at some stages of visual processing, there does appear to be selection directly by features, although it seems to be used less often and have fewer measurable effects than selection by location.

Attention to Objects

The evidence reviewed so far paints a picture of a spatial attention mechanism that selects locations and that can be guided by the presence of features such as color and orientation, augmented by a mechanism or mechanisms for selecting directly by those features without calling on spatial selection. There is another extensive body of evidence that reveals a different aspect of visual attention. In some tasks, spatial selection cannot easily separate a target from distractors, yet performance is sometimes very good, as was demonstrated early on by Neisser and Becklen (1975). They superimposed two different video clips. Their subjects could attend effectively to one clip while ignoring the other, even though both occupied the same general region in the display. In fact, as demonstrated more recently by Simons and Chabris (1999), the inhibition of the unattended information is so effective that very salient and surprising stimuli can be missed. Duncan (1984) kicked off the more controlled exploration of attention to a small target object with a superimposed distractor by asking subjects to report two properties from two simple geometric shapes. Performance was better when both properties were from the same shape, leading Duncan to conclude that attention could be allocated to one shape or the other, even though they occupied approximately the same location. When the two properties were from two different shapes, attention apparently had to shift from one object to the other, resulting in lower accuracy. Duncan's results could be explained by a very precise selection of location that could select only the contours of the target object while avoiding those of the distractor, but if spatial attention is as imprecise as suggested by Eriksen and Eriksen's (1974) flanker task and by Downing's (1988) demonstration of attentional gradients, then something else must underlie selection in Duncan's task. He suggested that in this task, attention was allocated to an object rather than a location.

Object-based attention has been studied with a variety of experimental procedures very different from Duncan's superimposed object paradigm (for reviews, see Chen, 2012; Scholl, 2001). The most popular was introduced by Egly, Driver, and Rafal (1994). The study was originally designed to explore attentional deficits in parietal patients, but their method has now been employed in many studies not involving brain damage. The method is an extension of the basic Posner cuing paradigm, in which two squares are presented, one on either side of fixation, and one of the squares is cued by making its contour brighter for a brief interval. A stimulus that appears in the cued square generates a faster response than a stimulus in the uncued square. This result was interpreted as evidence for location-based attention, but it could also be explained as the result of attention allocated to the cued object. To distinguish between these two possibilities, Egly, Driver, and Rafal extended the two boxes to make them long rectangles, both oriented in the same direction and one on each side of fixation. In their task, an entire rectangle was not cued; instead, the cue appeared only at the end of one of the rectangles. Thus, one of four locations could be cued (either end of either rectangle), and the probe stimulus could also appear at any of those four locations. As might be expected, responses to the probe were fastest when it appeared at the cued location. The key comparison in this experiment was between response times for probes at two other locations, both of which were equidistant from the cued location. One was on the same rectangle as the cue, but at the opposite end, and the other was at one end of the uncued rectangle. Responses to both of these locations were slower than responses to the cued location, but responses to the cued-object location were nonetheless faster than those to the uncued-object location. Because these two locations were equidistant from the cue, the difference could not be attributed to an attentional gradient. Instead, attention seemed to be stronger at all locations in the cued object than to locations in the uncued object. Attention seemed to spread to some extent to the entire object. The task in this experiment was to detect the probe regardless of which object it appeared in, and favoring locations in one object over locations in the other would not improve overall performance. Therefore, the result suggests that the attentional spread might occur automatically, regardless of the goals or the task. Avrahami (1999) demonstrated that these effects of stimulus organization did not require closed shapes.

Egely, Driver, and Rafal's object-based effect is very robust in the sense that it is easy to replicate, but it can be reduced or eliminated by manipulating a number of different factors. One important factor is attentional zoom. Goldsmith and Yeari (2003) demonstrated that the object boundaries only affect responses when attention is spread broadly across the display rather than narrowly focused. In Egely, Driver, and Rafal's original experiment, the cue was an exogenous cue, so subjects probably began each trial with attention distributed in order to detect the cue. The object effect disappeared in a study by Maquistan (1997) using endogenous cues, and Goldsmith and Yeari attributed this to the narrower attentional focus that Maquistan's subjects would have adopted to select the cue in the center of the display.

The role of attentional zoom is probably related to a finding by Shomstein and Yantis (2002) using measures of distractor interference within a cross-shaped configuration rather than the two parallel rectangles used by Egely, Driver, and Rafal. In their experiment, the object effect only arose when subjects did not know at the beginning of the trial where the target stimulus would appear. In experiments in which the target always appeared in the center of the display, the object effect disappeared. They concluded that the object effect required positional uncertainty, and they went on to claim that the effects were not the result of attention spreading to fill the boundaries of an object but instead the result of a process of prioritizing locations for a visual search. When the target location was not known in advance, a search was necessary to find it, and locations within the same object as the cue were given higher priority so that they would be more likely to be attended before locations outside the object.

Positional uncertainty does seem to be a relevant factor in the activation of object-based attention, perhaps because attention may be spread more broadly when the target location is unknown. However, it cannot be the only factor controlling object-based attention, because other studies have demonstrated object-based attention under positional certainty, including the Harms and Bundesen's (1983) study mentioned earlier, and another by Kramer and Jacobsen (1991). Chen and Cave (2006) used a stimulus in which the objects were in a cross configuration similar to that of Shomstein and Yantis. The object effect disappeared if the trials with the cross stimuli were mixed in with trials in which only one part of the complete configuration appeared. Because object-based attention can arise even with positional certainty, it does not seem to be simply a product of priority in visual search.

Lamy and Egeth (2002) proposed a different explanation of object-based attention. They concluded that object-based attention only arises after attention has shifted from one location to another. Their subjects compared two shapes that could appear either in the same rectangle or in different rectangles. The object effect only arose when one shape appeared after the other, allowing time for attention to select the first object and then shift to the second. Drummond and Shomstein (2010) argued that it was the positional uncertainty about the target locations in Lamy and Egeth's task rather than the attentional shift that triggered object-based attention.

Another factor that affects the object effect is the amount of time necessary to observe the stimulus. In a number of experiments (e.g., Avrahami, 1999; Chen & Cave, 2008; Law & Abrams, 2002; Shomstein & Behrmann, 2008), object effects become weaker or disappear if the stimulus is only available for a short time. These results suggest that organizing the scene into objects is not an obligatory step in the allocation of attention. Attention can be allocated spatially without the object representations playing a role. If more time is available, the object organization of the scene can be analyzed and used to shape attention.

A final factor that can affect object-based attention is the interpretation that the viewer imposes on the scene, as can be seen with stimuli that can be interpreted as either one object or multiple objects. Chen (1998) replaced Egely, Driver, and Rafal's two rectangles with a configuration made up of two V-shaped polygons joined at their points to form an X. When the configuration was described to subjects as two V's, the object effect was present, but when it was described as a single X, they disappeared. As mentioned earlier, Chen and Cave (2006) used a cross-shaped configuration. In this study, one group of subjects were

encouraged to interpret the configuration as separate shapes by exposing them to trials in which only part of the configuration was present. On the trials with the complete configuration, those subjects showed an object effect, even though another group of subjects showed no object effect with the same stimuli. Perhaps the object effect can be avoided by treating the entire configuration as a single object.

Object-based attention does not arise in all tasks, but it does arise in many tasks in which the object organization is irrelevant. Its presence is affected by several factors, but there does not appear to be a single factor that triggers it in all circumstances. Attention is not automatically shaped by object boundaries in every task, and object-based attention can appear even when a search is not necessary. Although there has been much disagreement about what underlies object-based attention, and even disagreement about the terminology to use, there does seem to be a general understanding now emerging that object-based attention is a very flexible mechanism that will be activated under some conditions and not others (see Drummond & Shomstein, 2010). More research is necessary to determine exactly what those conditions are, and why object-based attention is activated in some cases when it does not seem to contribute to the task.

In the end, the problem of explaining the control of object-based attention is turning out to be similar to the problem of explaining exactly how attention is allocated during visual searches. As with search, there is more flexibility in the control of object-based attention than was first appreciated, and that flexibility makes it difficult to arrive at a single unambiguous account of its operation.

The Relationship Between Spatial Attention and Object-Based Attention

If object-based attention is invoked in some circumstances, while location-based attention is left to work by itself in others, then what is the relationship between the two? They could be separate systems, operating at different levels of processing. As described earlier, the ventral or “What” visual pathway can be seen as a progression from spatially organized representations to representations organized by complex shape properties and ultimately to object representations. There could be selection mechanisms working throughout this hierarchy, with selection being more location based at the lower levels and more object based at the higher levels. At the highest level, object-based selection would be working among object representations that are coded independently of location, selecting some objects and/or inhibiting others.

Alternatively, there might be a single, more unified system. This system is ultimately spatial, selecting by location, but in many cases when there is enough time for building object representations, those representations can affect which locations are selected. Vecera and Farah (1994) referred to this location-mediated object selection as the “grouped array hypothesis.” In this account, selection is more of an interaction between spatial representations and object representations, and it probably requires some communication back and forth between different processing systems. This need for interaction between upper and lower levels arises in aspects of visual processing other than just the control of attention, and it has led to recurrent theories of visual processing (Di Lollo, Enns, & Rensink, 2000; Hochstein & Ahissar, 2002). When a stimulus first appears, there might be an initial wave of neural activity starting at the retina and working its way up through the LGN, V1, and the “What” pathway until high-level object representations are activated. Those object representations could then trigger another wave of activity back down the hierarchy to shape the allocation of attention at the lower, spatially organized levels.

Spatial Attention in Nonvisual Modalities

Spatial location is a fundamental property in visual processing, but it is also an important factor in audition and touch. Although the majority of recent research in attention has been in the visual domain, audition was the modality of choice for much of the foundational attentional research of the 1960s, and the results showed that attention for auditory stimuli can clearly be allocated to a specific location. Selection by location is also possible for touch stimuli. Furthermore, in many circumstances there are links between these different modalities, so that attention to the location of a stimulus in one modality can affect the processing of stimuli in the other modalities (Reisberg, Scheiber, & Potemken, 1981; Spence & Driver, 2004). One important question for future research is how information about location is shared across the different modalities to control attention. The ability to select multimodal information from a single location is probably related to the ability to attend to an object, and the two issues should be considered together.

Conclusion

Attention is necessary in vision to allocate higher level resources effectively and prevent interference between objects. An important aspect of visual attention is the selection of information from specific locations, and this spatial selection is apparently done within a three-dimensional mental representation. Spatial selection interacts with the presence of visual features, such as color and orientation, and also with boundaries and clusters that define objects and groups.

Future Directions

1. Are there separate mechanisms to select locations, features, and objects, or is there a single location-based mechanism that is driven by the presence of features and by object boundaries?
2. When can spatial attention be split across multiple locations simultaneously? What are the costs and benefits of splitting attention?
3. When attention is set to select information at a specific size or scale, is it selecting a specific region of that size, or does this scale selection work independently of location?
4. In visual search, how large a region is selected at any given moment? How does the size of the selected region vary with the relationship between target and distractors?

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